

DSA Animation Color Guide

UNIVERSAL COLOR ROLES

■ data structure ■ —

BLUE #1A73E8

Scanning / Processing

"[REDACTED], [REDACTED] [REDACTED]"

42



GREEN #22c55e

Success / Found / Correct

"[redacted], [redacted] [redacted]"

42



RED #ef4444

Error / Not Found / Wrong

"[REDACTED], [REDACTED] [REDACTED]"

42



YELLOW #eab308

Active Focus / Current Step

"[REDACTED] [REDACTED] [REDACTED] [REDACTED]"

42



ORANGE #f97316

Visited / Already Processed

[illegible]

42



WHITE #f9fafb

Static / Label / Neutral

"■■■■ ■■■■ information"

42



GRAY #6b7280

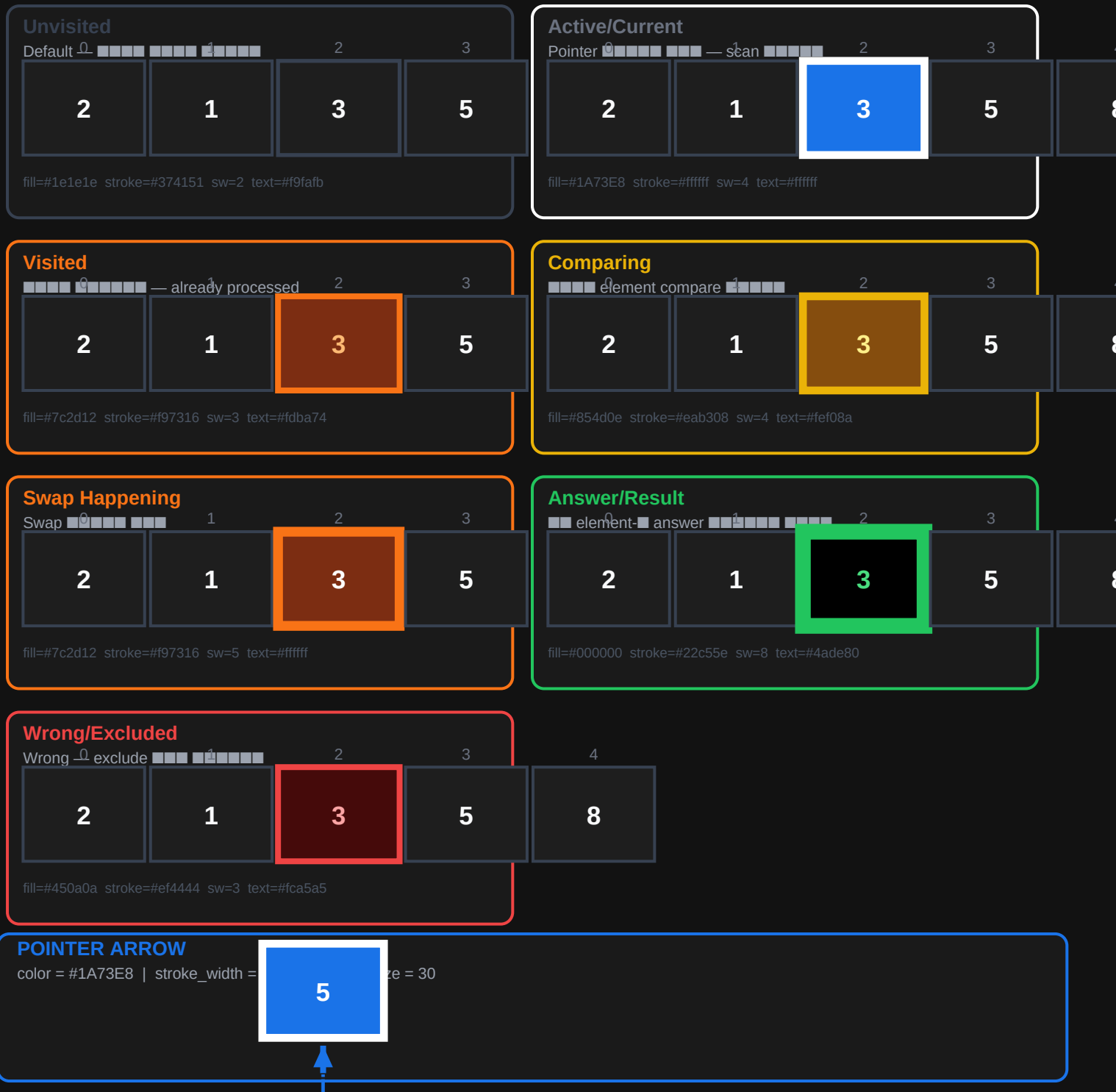
Inactive / Background

"■■■■ ■■■■ *relevant* ■■"

42

1. ARRAY

Linear structure — index access



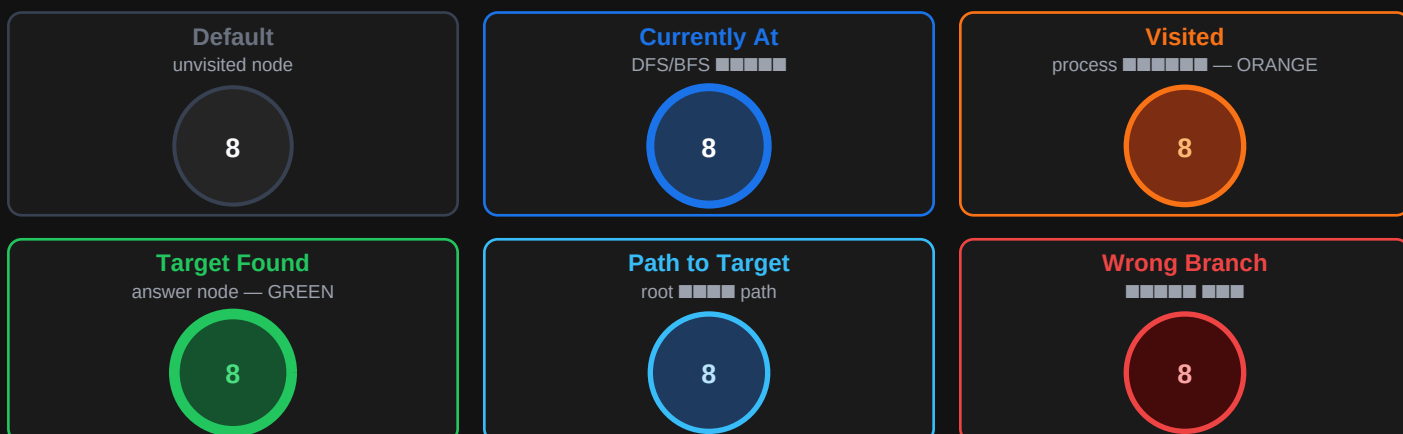
2. LINKED LIST

Node → next pointer chain



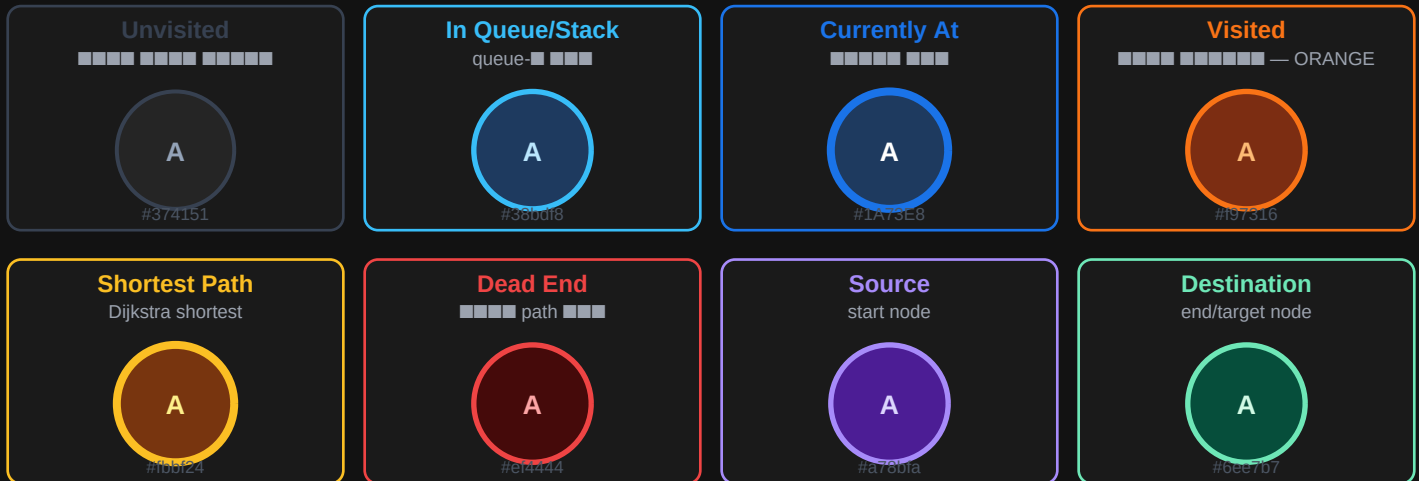
3. TREE

Binary Tree / BST — visited = ORANGE (not green!)



4. GRAPH

BFS / DFS / Dijkstra — visited = ORANGE



EDGE TYPES



5. HASH MAP

Key-Value table row states

KEY		VALUE / INDEX
7	0	Just Inserted notun add hoyeche
2	1	Being Scanned match khuja hochhe
1	3	Found (Match) complement pawa gache
5	2	Collision same bucket — collision

6. SET

Unique elements — circle/rounded visual

Empty Set

■■■■ element ■■■■

x

Element Present

■■■

x

Just Added

■■■■ insert — FadeIn+scale

x

Duplicate Hit

already ■■■■ — Shake

x

Found/Contains

■■■■■ ■■■■■■

x

Removed

delete — FadeOut

x

ANIMATION EFFECTS

■■■■ effect ■■■■ use ■■■■

Flash()

Answer found / success moment

flash_radius=1.0, num_lines=16, color=GREEN

Indicate()

Current element focus

color=BLUE

Wiggle/Shake

Duplicate / not found

scale_value=1.2, color=RED

FadeOut()

Remove / delete element

run_time=0.5

Flash()

Important highlight

flash_radius=1.5, num_lines=24, color=YELLOW

Circumscribe()

Final answer box

color=GREEN, fade_out=True

Blink stroke

Active scan border

width: 2→6→2, color=YELLOW

GrowFromCenter

New element insert

run_time=0.6

GLOBAL TEXT HIERARCHY

Title (top)

color=#ffffff font_size=28 weight=BOLD

Section label

color=#9ca3af font_size=22

Node value

color=#f9fafb font_size=26

Index / key

color=#6b7280 font_size=16

Status message

color=#eab308 font_size=20 (then RED or GREEN)

Muted note

color=#4b5563 font_size=14

GOLDEN RULES

1 ■■■■■■ ■■■■■■■■ ■■■■ bright color — ■■■■ ■■■■ ■■■■ confused ■■■■

2 ■■■■ color ■■■■■■■■ ■■■■ meaning — ■■■■ swap ■■■■ ■■

3 Yellow fill-■■■ ■■■■ text = BLACK, ■■■■ ■■ fill-■■■ ■■■■ = WHITE

4 Success = ■■■■■■■■ GREEN | Error = ■■■■■■■■ RED — ■■■■ ■■■■ ■■

5 Visited = ■■■■■■■■ ORANGE — GREEN ■■ (green = success only)

6 BG (#121212) ■■■■ ■■■■■■■■ ■■■■ ■■ — ■■ video-■■■ same

7 Active = ■■■■■■■■ ■■■■■■■■ | Inactive = ■■■■■■■■ ■■■■■■■■

8 Result box-■ ■■■■■■■■ Flash() ■■■■■■ ■■■■ ■■■■